

$\equiv_5$ (multiples of 5)

$$\equiv_5 \subseteq \mathbb{N} \times \mathbb{N}$$

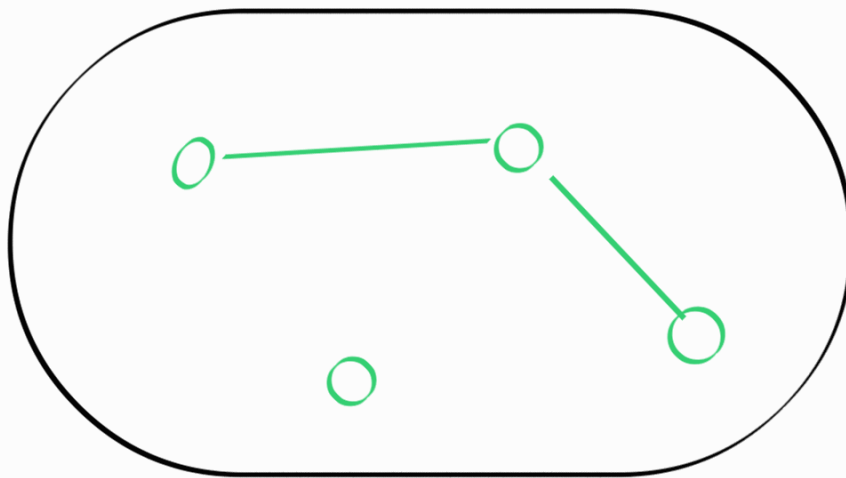
$$\equiv_5 = \{ \langle a, b \rangle \in \mathbb{N} \times \mathbb{N} : (a \bmod 5 = 0) \wedge (b \bmod 5 = 0) \}$$

1 2 3 4 5 ↻

Graphs

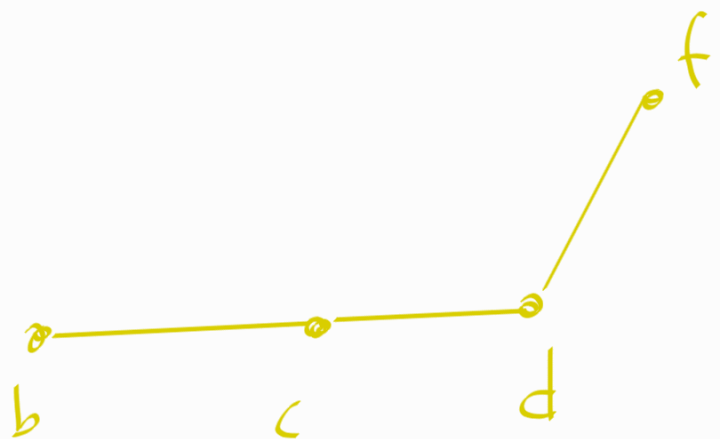
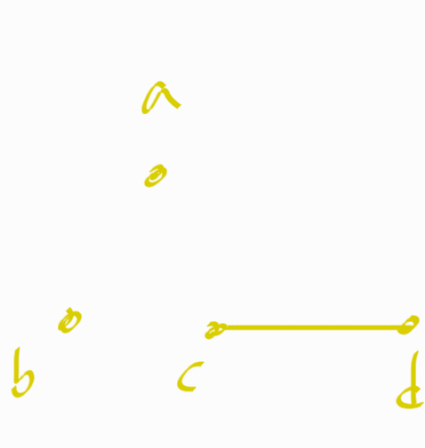
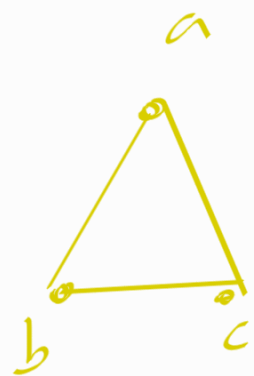
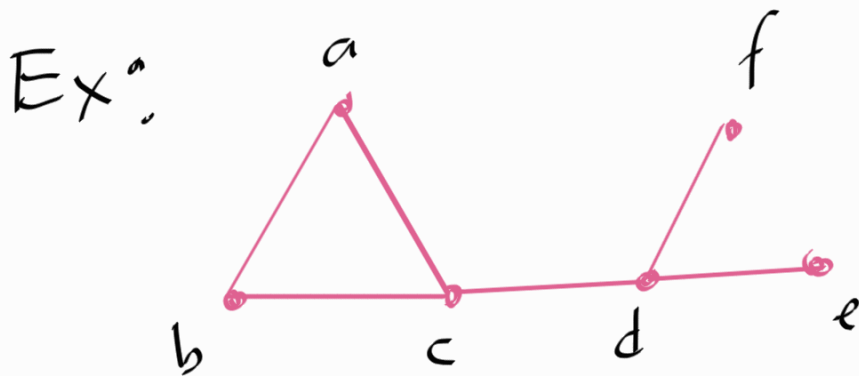
Simple graph

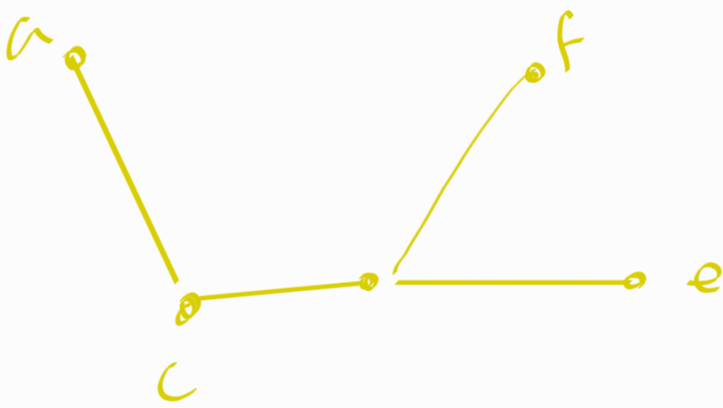
A graph with no self loops or parallel edges is called a simple graph.



Subgraphs

A graph G is a subgraph of graph H , If the nodes and edges of G are subsets of nodes and edges of H and node subset must include all endpoints of the edge subset.

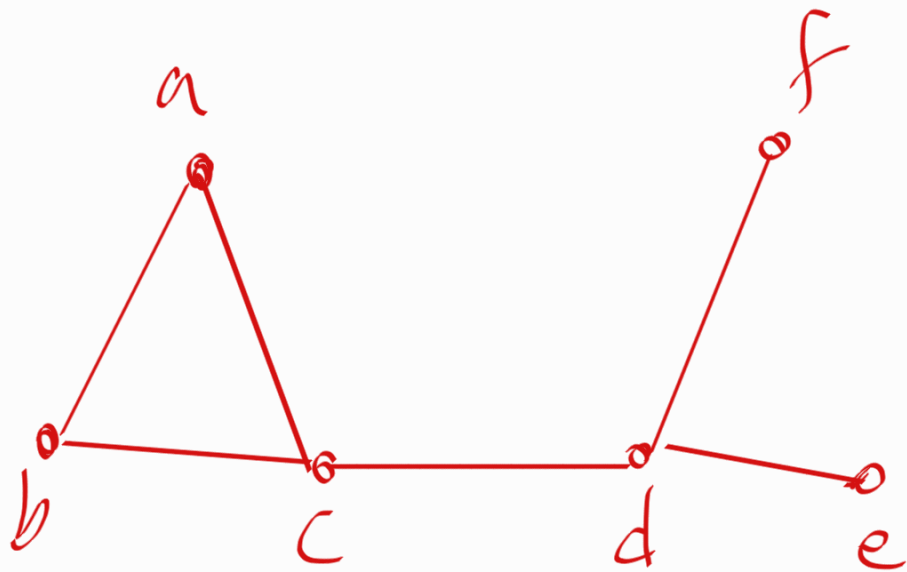




Enumerating all subgraphs could be costly.

- # of subgraphs is exponential

Connectedness: A path is a sequence of nodes connected by edges.



$\langle \underline{a}, \underline{b}, \underline{c}, \underline{d}, \underline{e} \rangle \checkmark$

$\langle a, b, c, e \rangle \times$

A simple path is a path that has no repeated vertices.

$\langle a, b, c \rangle$ simple path

(a, b, c, a, b, c, d) ; a path but not a simple path.

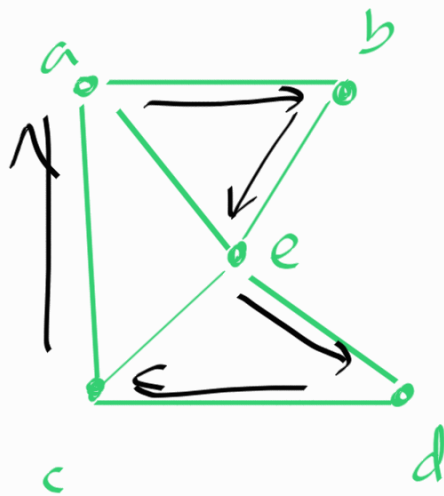
A path is a cycle if it starts and ends at the same node.

(a, b, c, a)

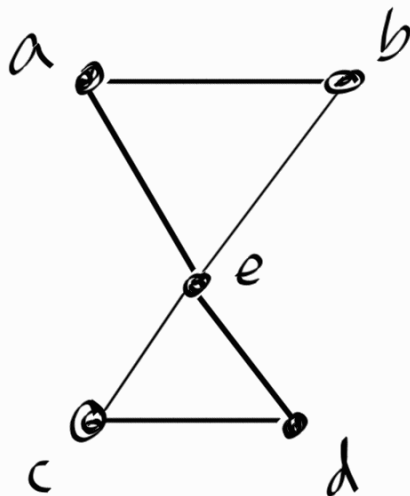
Hamiltonian cycle

A cycle that visits all nodes of the graph exactly once

Ex 1



Ex 2:



Induction

Mathematical induction is a method of proving predicates over natural numbers.

We can prove claims in the form of

$$[\forall n \in \mathbb{N} : P(n)]$$

$$P(1) \wedge P(2) \wedge P(3) \dots \wedge$$

1. The claim should be expressed in the form of predicates over natural numbers
2. Then we prove the predicate is true for the smallest natural number (Base case)
3. Then we prove the inductive step

$$\forall n \geq 1: p(n) \Rightarrow p(n+1)$$

4. We conclude that predicate is true for all natural numbers.

$$\sum_{i=1}^n i^2 = 1^2 + 2^2 + 3^2 + \dots + n^2$$

$$= \frac{1}{6} \cdot n(n+1)(2n+1)$$

Define predicate $P: \mathbb{N} \rightarrow \{\text{true}, \text{false}\}$

$$P(n) = \begin{cases} \text{true}, & \text{if } \sum_{i=1}^n i^2 = \frac{1}{6} \cdot n(n+1)(2n+1) \\ \text{false}, & \text{otherwise} \end{cases}$$

$$[\forall n \in \mathbb{N} : P(n)]$$

$$\sum_{i=1}^1 i^2 = \frac{1}{6} \cdot (1+1)(1)(2+1)$$

$$1^2 = \frac{1}{6} \cdot 2 \cdot 3 = 1$$

We want to prove

$$[\forall n \in \mathbb{N} : P(n)]$$

1. Show the base case is true.

$$n = 1$$

WTS: $P(1)$ is true.

$$P(1) \text{ basically says } \underbrace{\sum_{i=1}^1 i^2}_{\text{L.H.S}} = \underbrace{\frac{1}{6} \cdot 1 \times 2 \times 3}_{\text{R.H.S}}$$

$$\text{L.H.S} = 1^2 = 1$$

$$\text{R.H.S} = \frac{1}{6} \times 2 \times 3 = 1$$

$$\text{L.H.S} = \text{R.H.S}$$

$\therefore P(1)$ is true.

2. WTS: $\forall n \geq 1: P(n) \Rightarrow P(n+1)$

$$\forall n \geq 1: \sum_{i=1}^n i^2 = \frac{1}{6} \cdot n \cdot (n+1) (2n+1) \Rightarrow p(n+1)$$

$\underbrace{\hspace{10em}}_{p(n)}$

$$\hookrightarrow \underbrace{\sum_{i=1}^{n+1} i^2}_{L.H.S} = \frac{1}{6} \underbrace{(n+1)(n+2)(2 \cdot (n+1) + 1)}_{R.H.S}$$

Assume inductive hypothesis is true.

$p(n)$ is true.

WTS: $p(n+1)$

$$L.H.S = \sum_{i=1}^{n+1} i^2 = \sum_{i=1}^n i^2 + (n+1)^2$$

$$= \frac{1}{6} \cdot n(n+1) \cdot (2n+1) + (n+1)^2$$

$$= \frac{1}{6}(n+1) \left[2n^2 + n + 6n + 6 \right]$$

$$= \frac{1}{6}(n+1) [2n^2 + 7n + 6]$$

$$= \frac{1}{6}(n+1)(n+2)(2(n+1)+1)$$

$$\rightarrow [2n^2 + 4n + 3n + 6]$$

$$[2n(n+1) + 3(n+1)]$$

$$(n+1)(2n+3)$$

$$(n+1)(2n+2+1)$$

$$(n+1)(2(n+1)+1)$$

Alphabet: Any finite set

Each element is called a letter.

Ex: $\Sigma = \{0, 1\}$, $\Sigma_2 = \{\star, \Delta\}$

$\Sigma_3 = \{A, C, G, T\}$

String: A finite sequence of symbols.

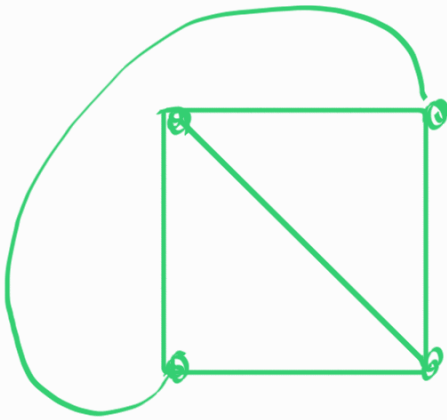
$S_1 = 0101$ $S_2 = 111100$ over
alphabet

$\Sigma = \{0, 1, 2\}$

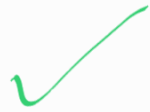
Empty string is denoted
using ϵ . $\leftarrow$ epsilon

Planar graphs

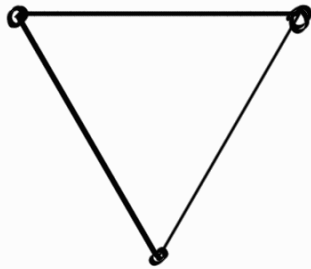
A graph whose vertices and edges can be drawn on the plane, with no edges crossing.



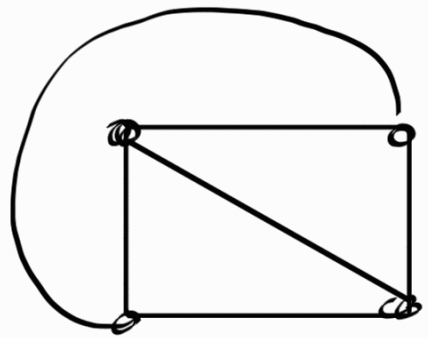
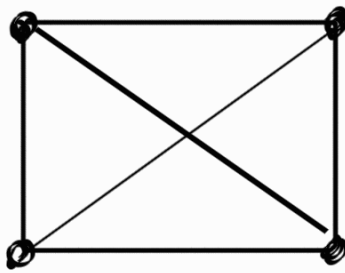
K_2 :



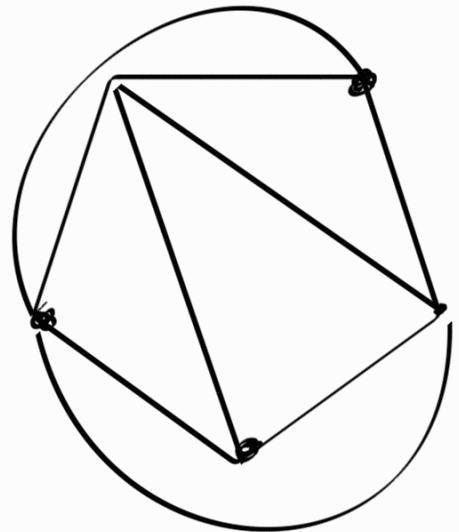
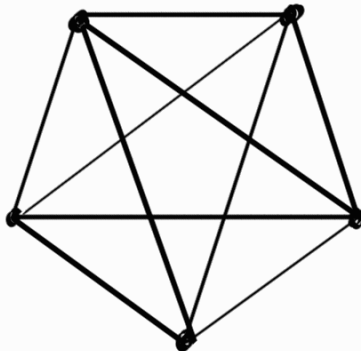
K_3 :



K_4 :

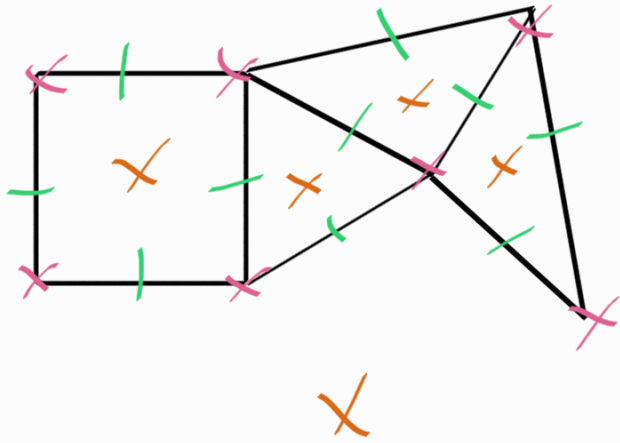


K_5 :



Euler's formula (for connected planar graphs)

$$|V| + |F| - |E| = 2$$



$$|V| = 7$$

$$|E| = 10$$

$$|F| = 5$$

In class exercise 01